Episcaf v3: designing from an energetic definition of the epitope

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Living record. Read top to bottom to get up to speed. Every methodological choice is a dated entry in `docs/DECISIONS.md`.

This is the running record of a new design avenue for episcaf, kept as its own subproject (`episcaf_v3/`) inside the episcaf repository. The shipped method at the repository root ("v2") scaffolds antibody epitopes defined by 3D *geometry*; this log asks whether defining the epitope by *energy* — which residues actually carry the binding — and letting the design constraints follow the energetics produces better scaffolds. It is built one recorded decision at a time (`docs/DECISIONS.md`) and is meant to be read in order: each section is one iteration of what we tried, what we found, and what it changed for the next step. It is not a formal manuscript, and the method here is early — most of it is still to build.

1 From geometric to energetic epitopes

1.1 What the previous method did

The shipped method ("v2"; it lives at the root of this repository, with `episcaf_v3/` as one subdirectory beside it) scaffolds *B-cell epitopes* — the regions of an antigen an antibody recognizes — so that a high-throughput serology assay (PepSeq) can test conformational, discontinuous epitopes that linear peptide tiling cannot represent. The epitope is taken from an antibody:antigen crystal structure as the set of antigen residues the antibody contacts, defined by a heavy-atom distance cutoff. Those residues are embedded into a *de novo* protein with RFdiffusion, given a sequence by ProteinMPNN, and folded by AlphaFold3 to check whether the design actually holds the epitope in its native geometry. Designs are kept or rejected on four filters: the epitope and overall backbone RMSD to the intended geometry, the predicted alignment error, and whether scaffold mass clashes into the antibody's binding path. Where an antigen has no solved antibody, a native-aware cylinder surrogate stands in for that clash filter. A composite scorer then ranks the survivors and picks the best designs per epitope. This pipeline produced DP4, a 36 000-member PepSeq library now in synthesis.

1.2 Where the geometric definition runs out

The method works, but two problems trace to the same root: it treats the epitope as a flat set of geometric contacts, all equally important.

The first is that binding energy is not uniform across an interface. A handful of "hot-spot" residues typically carry most of the binding free energy, surrounded by a ring of residues that contribute little directly but shape the pocket and exclude solvent. A geometric cutoff cannot see this: it labels every contacted residue an epitope residue, and the contig we hand RFdiffusion then fixes the identity and backbone of all of them equally. We are constraining residues we have no evidence are load-bearing, which costs the generator design freedom it may need to fold a good scaffold — and we do not actually know which contacts are carrying the binding.

The second is visible in the hard cases. A contig enforces epitope residue identity and ordering, but it does not strongly constrain the distances and orientations *between* spatially separated epitope islands. So the model can satisfy the contig and still lose the geometry an antibody needs. The clearest symptom is PDB `2XQB` patch `OP`, a 56-residue epitope split across two islands: across 2624 RFdiffusion trials, zero designs passed the filters. A 0% rate over thousands of attempts is a real failure of what the contig constrains, not undersampling.

Underneath both is a definitional gap. "Epitope" is loosely and inconsistently defined in the literature — usually by a geometric contact convention chosen for convenience — and that ambiguity propagates straight into the design constraints. If we cannot say which residues matter and why, we cannot say which ones a design is actually required to reproduce.

1.3 The new avenue: define the epitope by energy

v3 replaces the geometric definition with an energetic one. Instead of asking which residues *touch* the antibody, we ask which residues *carry the binding*, and we let the design constraints follow the energetics. Concretely, we sort the interface into three categories and constrain each according to what its job is:

- **Category 1 — energetic (hot-spot) residues.** The residues that dominate the binding free energy. These retain both their *exact identity and their shape*: the contig fixes their backbone and their sequence is locked to native. This is what we currently do to the whole geometric epitope; here we do it only where it is load-bearing.
- **Category 2 — structural (occluding) residues.** The rest of the contact shell. These matter for how the site fits together — they hold the shape and exclude solvent around the hot spots — but their identity is not the point. So we fix their *shape* (the contig holds the backbone) but let ProteinMPNN choose the sequence. Their correct orientation can be enforced by superimposing them on the native antigen and filtering on a small RMSD.
- **Category 3 — the rest of the scaffold.** These residues exist to enforce the geometry of Categories 1 and 2. They can be any identity and adopt any structure, subject to two constraints: they must not introduce new steric clashes with the antibody, and they must not compete with the binding site. Everything else about them is free — scaffold length and the position of the epitope within the construct are randomized.

The bet is simple: by fixing identity only where it carries the binding, and shape only where a residue's job is structural, we give the generator far more room to build a foldable scaffold without giving up the interaction that matters. The same energetic reading also sharpens the hard-case problem — it tells us which inter-residue geometry is worth enforcing rather than enforcing all of it blindly.

A second change makes the pipeline cheaper and is, in hindsight, overdue: we can filter RFdiffusion backbones *before* ProteinMPNN and AlphaFold3. A backbone that already places the epitope in the wrong orientation, or drapes scaffold mass through the antibody's path, was never going to produce a passing design; superimposing the fixed epitope on the native antigen and checking a small RMSD plus an antibody-clash test at the backbone stage discards those doomed designs before we spend the expensive sequence-design and folding steps on them. This is the same accessibility-and-fidelity check the shipped scorer applies at the end, moved to the front.

1.4 What this rests on

The whole approach stands or falls on getting the energetics right, and the hard part is solvent. A naive pairwise antigen–antibody interaction energy will mislead: it ignores the desolvation a residue pays to make a contact, and it misses contributions mediated by ordered water — a bridging hydrogen bond can be load-bearing yet invisible to a direct pairwise sum. Deciding how to compute a defensible per-residue energy, including solvent, is the first real decision of v3, and it is where the record begins.

We will not trust the ranking on faith. The first step is to reproduce known numbers: take a complex with experimentally measured binding energetics and check that our per-residue energies recover its known hot spots before we let the definition drive any design. The DP4 binding data, when it returns, gives a second, independent test — whether designs that better preserve the energetic residues bind better.

We are not starting the energy machinery from scratch. The sibling `bcell_epitope` project already computes a per-residue interface interaction-energy decomposition — including a water-mediated bridge channel — on a documented GROMACS stack, and it is the template we build the energy engine from. We treat it as a reference, not a protocol to inherit: its "epitopeness" is an interaction energy, deliberately not a binding free energy (it omits desolvation and entropy), so whether v3 adopts that quantity or a fuller one is itself a decision to make and record, not to assume.

1.5 Status

This is the beginning. The method above is a plan, not a result: nothing here has been built or tested yet. What follows in this log will be the decisions, one at a time — the energy method, the MD protocol, the classification thresholds, the contig strategy, the filters — each with the options weighed, the choice made, and the check that backs it, mirrored in `docs/DECISIONS.md`.

References